

Manufacturing Operations Data

Operation	Material	Machine	Duty	Shape	Setup Time (min)	Cycle Time (sec)	Work Hours	Diameter (mm)	Length (mm)
Drilling	Steel	Drilling machine	Heavy duty	Round	25	120	1800	22	65
Turning	Aluminium	CNC	Medium duty	Round	30	90	2400	40	120
Milling	Steel	3 axis CNC	Heavy duty	Rectangular	45	150	3600	60	200
Grinding	Steel	SPM	Heavy duty	Round	35	180	4000	25	80
JIG boring	Titanium	5 axis CNC	Medium duty	Round	50	210	3200	18	90
Rubber press	Steel	Rubber press - Small	Heavy duty	Square	40	300	2800	150	400
Rubber press	Aluminium	Rubber press - Medium	Medium duty	Rectangular	55	280	3500	220	600
Drilling	Aluminium	Conventional	Light duty	Round	15	60	1200	12	50
Turning	Steel	CNC	Heavy duty	Round	28	110	2600	55	180
Milling	Titanium	5 axis CNC	Heavy duty	Rectangular	60	240	4200	70	300
Grinding	Steel	SPM	Medium duty	Round	32	160	2900	30	100
JIG boring	Steel	3 axis CNC	Heavy duty	Round	48	200	3300	20	85
Rubber press	Steel	Rubber press - Medium	Heavy duty	Rectangular	58	320	4100	250	750
Drilling	Steel	Drilling machine	Medium duty	Round	22	95	1900	18	60

Operation	Material	Machine	Duty	Shape	Setup Time (min)	Cycle Time (sec)	Work Hours	Diameter (mm)	Length (mm)
Turning	Aluminium	Conventional	Light duty	Round	18	70	1500	35	140
Milling	Steel	CNC	Medium duty	Square	40	130	2700	45	160
Grinding	Titanium	SPM	Heavy duty	Round	38	190	3600	28	95
JIG boring	Steel	5 axis CNC	Heavy duty	Round	52	230	3900	24	110
Rubber press	Steel	Rubber press - Small	Medium duty	Square	42	260	3000	180	500
Rubber press	Aluminium	Rubber press - Medium	Heavy duty	Rectangular	60	340	4500	300	900